



Assignment 01

Instructor: **Mr. Saifullah Tanvir**

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Question no 1: Spotify Homepage Clone

Total Marks: **50**

Create a clone of the Spotify Web Player homepage. Everything that appears on the original website should be included in your clone.

Requirements:

1. The design must be fully responsive (mobile, tablet, desktop).
 - You may use tools like Responsively to check and improve your responsive design.
2. Add a Light/Dark mode toggle button in the navbar that switches the background and text colors of the page.
3. **Do not use** any AI tools, Copilot, or Cursor for generating code.
 - In evaluation, each student will be asked relevant questions about their own code and implementation.
 - Marks will be based on this evaluation.
 - If a student fails to explain their work or code, they will be awarded **zero marks**.
 - Sample images are attached at the end of the assignment for your reference.

Question no 2:

Write a JavaScript function **processArray(arr, callback)** that takes an array of numbers and a callback function. The function should apply the callback to each element of the array and return a new array with the results.

- If the callback is double, each value in the array should be doubled.
- If the callback is square, each value in the array should be squared.

```
const numbers = [10, 5, 8, 2, 6];
```

```
processArray(numbers, double);  
// Output: [20, 10, 16, 4, 12]
```

```
processArray(numbers, square);  
// Output: [100, 25, 64, 4, 36]
```

Question no 3:

You are given the following array of student objects:

```
const students = [  
  { name: "Ali", age: 22, marks: 85 },  
  { name: "Ayesha", age: 20, marks: 92 },  
  { name: "Bilal", age: 23, marks: 78 },  
  { name: "Ibrahim", age: 22, marks: 76 },  
  { name: "Usama", age: 21, marks: 80 }  
];
```

Write JavaScript functions to:

1. Sort the students by marks in descending order.
2. Return the names of students who have marks > 80.
3. Return the student objects of those who are older than 21.

Question no 4:

You are given the following HTML structure:

```
<input type="text" id="nameInput" placeholder="Enter name">  
  <button id="addButton">Add Name</button>  
  <ul id="nameList"></ul>
```

Write JavaScript that performs the following:

1. When the button is clicked, the entered name should be added as a new inside the .
2. Prevent empty inputs from being added to the list.

Special Note: I want you to genuinely learn something from this course, so don't assume that I am from your batch and will be doing any kind of special favor for you. The assignment is straightforward and easy if you put in the effort.

If you have any query, you can reach me at I227992@lhr.nu.edu.pk.

And if anyone is curious about who I am or what I do, feel free to check out my LinkedIn: [Wasee-Ur-Rehman](#)



